

MQTT: The Standard for IoT Messaging

MQTT is an OASIS standard messaging protocol for the Internet of Things (IoT). It is designed as an extremely lightweight publish/subscribe messaging transport that is ideal for connecting remote devices with a small code footprint and minimal network bandwidth. MQTT today is used in a wide variety of industries, such as automotive, manufacturing, telecommunications, oil and gas, etc.

Getting started with MQTT

Why MQTT?

Lightweight and Efficient

MQTT clients are very small, require minimal resources so can be used on small microcontrollers. MQTT message headers are small to optimize network bandwidth.

Bi-directional Communications

MQTT allows for messaging between device to cloud and cloud to device. This makes for easy broadcasting messages to groups of things.

Scale to Millions of Things

MQTT can scale to connect with millions of IoT devices.

Reliable Message Delivery

Reliability of message delivery is important for many IoT use cases. This is why MQTT has 3 defined quality of service levels: 0 - at most once, 1- at least once, 2 - exactly once

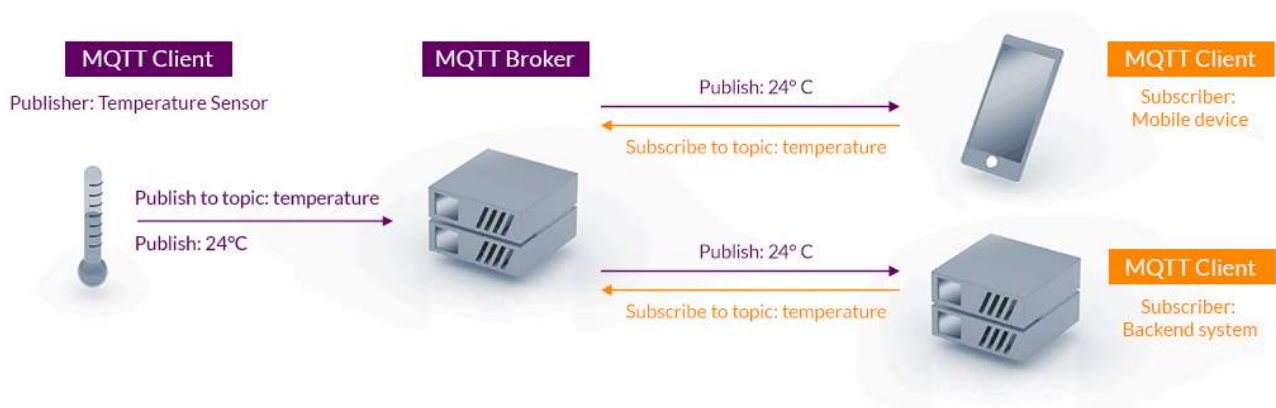
Support for Unreliable Networks

Many IoT devices connect over unreliable cellular networks. MQTT's support for persistent sessions reduces the time to reconnect the client with the broker.

Security Enabled

MQTT makes it easy to encrypt messages using TLS and authenticate clients using modern authentication protocols, such as OAuth.

MQTT Publish / Subscribe Architecture



MQTT in Action

MQTT is used in a wide variety of industries

Automotive

Logistics

Manufacturing

Smart Home

Consumer Products

Transportation

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